Peel and Glue: The Local Rule Operations on Cylindric Growth Diagrams

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Abstract

We give a self-contained account of the *peel* and *glue* operations on cylindric growth diagrams. These are the inverse and forward Burge local rules, applied one full ring at a time around the cylinder. They provide the mechanism for the sign-reversing involution on the ground set $\widetilde{\mathcal{G}}_{n,c}$ of triples (S, U, ν), where S records crossing levels, U records Euler marks, and ν is a cylindric plane partition.

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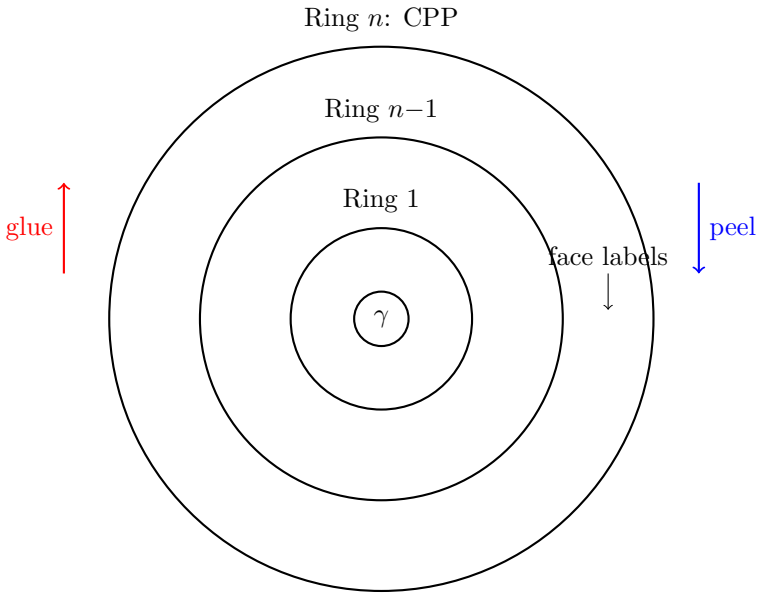
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1 The growth diagram as an onion

A cylindric plane partition ν with max entry n lives on the outer boundary of a cylindric growth diagram. The growth diagram has n concentric *rings*, with a base partition γ at the centre.



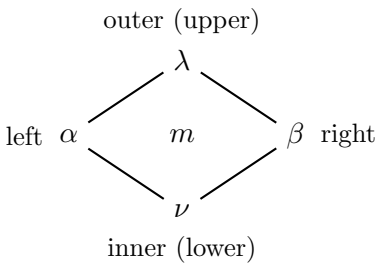
At each ring, the growth diagram carries two types of data:

- **Vertex labels**: partitions at each vertex around the ring. The outermost ring gives the CPP $\nu = (\mu^0, \mu^1, \ldots, \mu^{T-1})$. The innermost vertex labels are all equal to the base partition γ.
- **Face labels**: a non-negative integer $m(f) \geq 0$ at each face between two consecutive rings. These form the ALCD (arbitrarily labelled cylindric diagram).

The two operations—**peel** (move inward) and **glue** (move outward)—transfer data between the vertex labels and the face labels, one ring at a time.

2 The Burge local rule

The local rule operates on a single *face* of the growth diagram: a diamond with four vertices.



The four vertex labels are partitions satisfying: α/ν and β/ν are horizontal strips (going outward), and λ/α and λ/β are horizontal strips (going outward). The face label $m \geq 0$ mediates between the inner and outer vertices.

Remark 1 (The commutation relation)**.** The local rule is the bijective counterpart of the commutation relation between vertex operators:

$$T(v)\,P(w) = \frac{1}{1-vw}\,P(w)\,T(v),$$

where T is the transfer (creation) operator and P is the projection (annihilation) operator on the space of partitions. Each application of the local rule at one face corresponds to one commutation step: swapping a T past a P. The scalar $1/(1-vw) = \sum_{m\geq 0}(vw)^m$ generates the face labels—the face label m records which term in the expansion we are in. The forward rule (glue) corresponds to the left-hand side $T\,P$; the inverse rule (peel) corresponds to the right-hand side $P\,T$.

2.1 The forward rule (glue direction)

Definition 2 (Burge forward rule)**.** Given inner vertex ν, side vertices α, β (with α/ν and β/ν horizontal strips), and face label $m \geq 0$, the outer vertex $\lambda = \mathfrak{U}_{\alpha,\beta}(m,\nu)$ is constructed as follows.

1. Work in **conjugate coordinates**: let α', β', ν' be the conjugate partitions.
2. Compute the **column sets**:

$$A := \{i \geq 1 : \alpha'_i > \nu'_i\} \qquad \text{(columns where } \alpha \text{ exceeds } \nu\text{)},$$
$$B := \{i \geq 1 : \beta'_i > \nu'_i\} \qquad \text{(columns where } \beta \text{ exceeds } \nu\text{)}.$$

   Since α/ν and β/ν are horizontal strips, $\alpha'_i - \nu'_i \in \{0,1\}$ for all i, so A and B are finite sets of positive integers.
3. **Cascade**: for each $i \in A \cap B$ (in increasing order), let $\varepsilon(i)$ be the smallest integer $> i$ not in $A \cup B$ and not previously assigned. Set $C := \{\varepsilon(i) : i \in A \cap B\}$.
4. **Free columns**: let D be the first m elements of $\mathbb{Z}_{>0} \setminus (A \cup B \cup C)$, in increasing order.
5. **Build** λ: set

$$\lambda'_i = \nu'_i + [i \in A \cup B \cup C \cup D].$$

The cascade step is the heart of the local rule. When columns i belongs to both A and B (i.e., both α and β exceed ν in that column), the column cannot simply be "used twice." Instead, it *cascades*: the overlap at column i pushes the activation to the next available column $\varepsilon(i) > i$.

Proposition 3 (Weight identity)**.**

$$|\lambda| = |\alpha| + |\beta| - |\nu| + m. \tag{1}$$

The face label m records the excess weight*: the amount by which λ exceeds what α and β alone would determine.*

Proof. We have $|A\cup B\cup C\cup D| = |A|+|B|-|A\cap B|+|C|+m = |A|+|B|+m$ (since $|C| = |A\cap B|$). And $|\lambda| - |\nu| = |A \cup B \cup C \cup D|$, $|\alpha| - |\nu| = |A|$, $|\beta| - |\nu| = |B|$. □

2.2 The inverse rule (peel direction)

Definition 4 (Burge inverse rule)**.** Given side vertices α, β and outer vertex λ (with λ/α and λ/β horizontal strips), the inverse rule recovers $(m, \nu) = \mathfrak{U}^{-1}_{\alpha,\beta}(\lambda)$:

1. Work in conjugate coordinates.
2. Compute the **excess column sets**:
$$\bar{A} := \{i \geq 1 : \lambda'_i > \alpha'_i\},$$
$$\bar{B} := \{i \geq 1 : \lambda'_i > \beta'_i\}.$$
3. **Reverse cascade**: for each $i \in \bar{A} \cap \bar{B}$ (in *decreasing* order), let $\delta(i)$ be the largest integer $< i$ not in $\bar{A} \cup \bar{B}$ and not previously assigned.
4. **Face label**: $m := \#\{i \in \bar{A} \cap \bar{B} : \delta(i) \leq 0\}$ (cascade chains that "fall off the edge").
5. **Build** ν: set
$$\nu'_i = \lambda'_i - [i \in \bar{A} \cup \bar{B} \cup \bar{C}],$$
where $\bar{C} = \{\delta(i) : \delta(i) > 0\}$.

The reverse cascade "unwinds" the forward cascade: each overlap column $i \in \bar{A} \cap \bar{B}$ traces back to a smaller column $\delta(i) < i$. If the trace reaches column 0 or below, it contributes to the face label m instead.

Proposition 5. *The forward and inverse rules are mutual inverses:* $\mathfrak{U}^{-1}_{\alpha,\beta}(\mathfrak{U}_{\alpha,\beta}(m, \nu)) = (m, \nu)$.

2.3 The cascade parameter in NILP terms

In the non-intersecting lattice path (NILP) model, the column sets A and B correspond to up-steps on two adjacent paths:

- A: columns where the left path p^α takes an up-step beyond the inner path p^ν.
- B: columns where the right path p^β takes an up-step beyond p^ν.

The cascade columns $A \cap B$ are where *both* adjacent paths rise simultaneously—a "pinch point" where the gap between paths narrows. The cascade resolution spreads these pinch points to neighbouring columns, preventing the paths from crossing.

The face label m counts the number of cascade chains that propagate all the way to column 0. In NILP terms, these are *cubes*—unit cells between the paths that contribute to the total weight.

$$\boxed{\text{face label } m = \text{number of cubes at this face} = \text{cascade chains reaching column 0.}} \tag{2}$$

3 Peeling a full ring

The growth diagram for a CPP of profile c with period $T = d$ has T faces per ring. Crucially, these T faces form a *closed circle*: face $T - 1$ connects back to face 0, reflecting the cylindric boundary conditions. Peeling a full ring means simultaneously applying the inverse local rule at all T faces around this circle, moving from ring k to ring $k - 1$.

In terms of the commutation relation (Remark 1), a full-ring peel commutes one transfer operator past a full cycle of T projection operators around the cylinder. Each face contributes one commutation step and one face label; the complete circle of T commutations produces the face-label sequence $(m_0, \ldots, m_{T-1})$.

3.1 The peel operation

Definition 6 (Full-ring peel)**.** Let $\nu = (\mu^0, \mu^1, \ldots, \mu^{T-1})$ be a CPP with max entry n. The **peel** at ring n produces:

1. A sequence of face labels $(m_0, m_1, \ldots, m_{T-1})$ at ring n.
2. A smaller CPP $\nu' = (\mu'^0, \mu'^1, \ldots, \mu'^{T-1})$ with max entry $\leq n-1$.

The peel is computed by applying the inverse Burge rule at each face $f_0, f_1, \ldots, f_{T-1}$ around the ring, using the profile π to determine which vertex is α and which is β at each face.

Proposition 7 (Peel weight transfer)**.** *The total weight transferred from the CPP to the face labels is:*

$$|\nu| - |\nu'| = \sum_{k=0}^{T-1} m_k. \tag{3}$$

Each face label m_k records the number of cubes at face k of ring n.

Remark 8 (Iterating the peel)**.** Applying the peel n times (once per ring, from ring n down to ring 1) completely decomposes the CPP into:

- The base partition γ (the core).
- The face labels at all n rings: an ALCD $\mathfrak{d} = (m_k^{(j)})$ with $j = 1, \ldots, n$ and $k = 0, \ldots, T-1$.

This is the growth diagram bijection $\psi\colon (\gamma, \mathfrak{d}) \leftrightarrow \nu$ of [1].

3.2 The glue operation

Definition 9 (Full-ring glue)**.** Given a CPP ν' with max entry $\leq n-1$ and a sequence of face labels $(m_0, \ldots, m_{T-1})$ with $m_k \geq 0$, the **glue** at ring n produces a CPP ν with max entry $\leq n$, by applying the forward Burge rule at each face around the ring.

Glue is the exact inverse of peel: peeling ring n and then gluing it back (with the same face labels) returns the original CPP.

3.3 What peel and glue look like

Example 10 (Rank 3, $c = (2, 2, 0)$, $d = 4$)**.** Consider the CPP $\nu = ((1, 1), (1, 1, 1), (1))$ with max entry 1 and weight 6. The profile gives a cylinder of period $T = 4$.

Peeling ring 1 applies the inverse rule at each of the 4 faces. The result is:

- Face labels: (m_0, m_1, m_2, m_3) — recording the cubes at each face of ring 1.
- Inner CPP: the base partition γ (since max entry was 1, one peel reaches the core).

The total face-label weight $\sum m_k = |\nu| - |\gamma|$.

4 The ground set and the involution

4.1 The three-component ground set

We begin from two definitions. The generating function for cylindric plane partitions with profile c is

$$\mathrm{GK}_c(z,q) = \sum_{\pi \in \mathrm{CPP}(c)} z^{\max(\pi)}\, q^{|\pi|}, \tag{4}$$

where $\max(\pi)$ is the largest entry and $|\pi|$ is the sum of all entries. The *Q-polynomial* is

$$Q_{n,c}(q) = (q;q)_n \cdot [z^n]\big((zq;q)_\infty \cdot \mathrm{GK}_c(z,q)\big). \tag{5}$$

We now expand (5) to derive the ground set. Write $g_m(q) = [z^m]\ \mathrm{GK}_c(z,q)$, so that $\mathrm{GK}_c = \sum_m g_m\, z^m$. The Euler product $(zq;q)_\infty = \prod_{k\geq 1}(1-zq^k)$ expands as

$$(zq;q)_\infty = \sum_J (-1)^{|J|}\, z^{|J|}\, q^{\sigma(J)},$$

where J ranges over all finite subsets of $\{1,2,3,\ldots\}$ and $\sigma(J) = \sum_{k\in J} k$. Multiplying by GK_c and extracting $[z^n]$ (which forces $|J| + m = n$, so $m = n - |J|$ and $|J| \leq n$):

$$f_n(q) := [z^n]\big((zq;q)_\infty \cdot \mathrm{GK}_c\big) = \sum_{j=0}^{n} (-1)^j \left(\sum_{\substack{J\subseteq\{1,2,\ldots\} \\ |J|=j}} q^{\sigma(J)} \right) g_{n-j}.$$

The sum over j-element subsets of $\{1,2,\ldots\}$ equals $q^{j(j+1)/2}/(q)_j$ (the minimal such subset $\{1,\ldots,j\}$ has sum $j(j+1)/2$; the excess is an unrestricted partition into at most j parts). So $f_n = \sum_j (-1)^j\, q^{j(j+1)/2}\, g_{n-j}/(q)_j$.

Multiplying by $(q)_n$ and splitting $(q)_n/(q)_j = \begin{bmatrix} n \\ j \end{bmatrix} \cdot (q)_{n-j}$:

$$Q_n = \sum_{j=0}^{n} (-1)^j\, q^{j(j+1)/2} \begin{bmatrix} n \\ j \end{bmatrix} (q)_{n-j}\, g_{n-j}.$$

Now use the identity $q^{j(j+1)/2} \begin{bmatrix} n \\ j \end{bmatrix} = \sum_{\substack{S\subseteq\{1,\ldots,n\} \\ |S|=j}} q^{\sigma(S)}$ (the minimal j-element subset of $\{1,\ldots,n\}$ is $\{1,\ldots,j\}$; the q-binomial counts the excess), and expand $(q)_m = \prod_{k=1}^{m}(1-q^k) = \sum_{U\subseteq\{1,\ldots,m\}} (-1)^{|U|}\, q^{\sigma(U)}$. Setting $m = n - j$:

$$\boxed{Q_{n,c}(q) = \sum_{(S,U,\nu)\in\widetilde{\mathcal{G}}_{n,c}} (-1)^{|S|+|U|}\, q^{\sigma(S)+\sigma(U)+|\nu|},} \tag{6}$$

where the **ground set** is

$$\widetilde{\mathcal{G}}_{n,c} = \bigsqcup_{j=0}^{n} \left\{ (S,U,\nu):\ \begin{array}{l} S\subseteq\{1,\ldots,n\},\ |S|=j, \\ U\subseteq\{1,\ldots,n-j\}, \\ \nu\in\mathrm{CPP}(c),\ \max(\nu)=n-j \end{array} \right\}. \tag{7}$$

Each component of the triple arises from a different factor in (5):

Component	Source	Sign	Weight
$S \subseteq \{1, \ldots, n\}$	$q^{j(j+1)/2} \begin{bmatrix} n \\ j \end{bmatrix}$ from $(q)_n/(q)_j$	$(-1)^{\lvert S\rvert}$	$\sigma(S) = \sum_{s \in S} s$
$U \subseteq \{1, \ldots, m\}$	$(q)_m$ factor	$(-1)^{\lvert U\rvert}$	$\sigma(U) = \sum_{u \in U} u$
$\nu \in \mathrm{CPP}(c)$, $\max = m$	$g_m = [z^m]\ \mathrm{GK}_c$	$+1$	$\lvert\nu\rvert$

Here $m = n - |S|$ is the max entry of the CPP. The larger the crossing set S, the fewer rings the CPP has: a triple with $|S| = j$ uses a CPP with only $n - j$ rings in the growth diagram. At one extreme ($S = \emptyset$), the CPP occupies all n rings; at the other ($|S| = n$), the CPP is empty and all weight is carried by S. The sign is $(-1)^{|S|+|U|}$ and the weight is $\sigma(S) + \sigma(U) + |\nu|$. All three components are manifestly non-negative objects (subsets and CPPs); the signs come *only* from the exponent $(-1)^{|S|+|U|}$.

Remark 11 (S as crossing levels)**.** The triple (S, U, ν) allocates n potential levels into two roles: the $m = n - |S|$ levels *not* in S are occupied by the CPP (one ring per level), while the $|S|$ levels in S are spent as *crossings*. Each element $k \in S$ is the index of a level allocated to crossings rather than to the CPP, contributing weight q^k and sign -1.

One might hope that k equals the number of boxes at the corresponding ring (the face-label total at that level). But this cannot work in general: two different rings could have the same box count, violating the requirement that S is a *set* of distinct elements. The box count could also be 0 or exceed n. So the elements of S are abstract level indices from the q-binomial expansion, not face-label totals.

In the involution, when we peel a ring and add an element to S, we must choose $k \in \{1, \ldots, n\} \setminus S$ and adjust U so that weight conservation holds. The face-label total at the peeled ring constrains this choice but does not determine it uniquely—this is part of the canonical choice (Section 4.5).

Remark 12 (U as compressed rings)**.** The subset $U \subseteq \{1, \ldots, m\}$ plays an analogous but subtler role to S. It comes from expanding the factor $(q)_m = \prod_{k=1}^{m}(1 - q^k)$: at each ring k, we choose either the 1 term (ring k is uncompressed) or the $-q^k$ term (ring k is compressed, contributing weight k and sign -1).

The element $i \in U$ means: *ring i has been compressed*. The ring is still present in the CPP—unlike S, where the ring is destroyed entirely—but i units of its face-label weight have been extracted and stored in U. The ring is lighter than it would otherwise be.

In this picture, a triple (S, U, ν) represents a single CPP that has been compressed in **two stages**:

1. **Structural compression** ($\to S$): interior-crush (Section 4.6) the levels in S out of the NILP tower. Each $k \in S$ removes level k, collapsing two adjacent levels into one. The CPP goes from max entry n to max entry $m = n - |S|$.

2. **Weight compression** ($\to U$): at each remaining ring $i \in U$, peel out i units of face-label weight. The ring stays but gets lighter.

What remains after both stages is ν. To **reconstruct** the full CPP, reverse the process: first un-peel the rings of U (inject weight back), then un-crush the levels of S (glue the removed levels back in).

The sign $(-1)^{|S|+|U|}$ counts the total number of compressions: $|S|$ structural crushes plus $|U|$ weight extractions. The involution toggles one compression—either crushing or uncrushing one more level (changing $|S|$), or peeling or unpeeling one more ring (changing $|U|$)—thereby changing $|S| + |U|$ by ± 1.

Remark 13 (Multiple compressions of a single CPP)**.** A single CPP $\tilde{\nu}$ with max entry n can be compressed into many different triples, depending on which levels are crushed (choice of S) and which rings have weight extracted (choice of U). For instance, with max entry 3, some of the triples representing different compressions of $\tilde{\nu}$:

Triple	**Compressions**	**Sign**
$(\emptyset,\ \emptyset,\ \tilde{\nu})$	none	$+1$
$(\{3\},\ \emptyset,\ \nu')$	crush level 3	-1
$(\{2\},\ \emptyset,\ \nu'')$	crush level 2	-1
$(\emptyset,\ \{1\},\ \nu''')$	peel ring 1	-1
$(\{2,3\},\ \emptyset,\ \nu^{(4)})$	crush levels 2 and 3	$+1$
$(\{3\},\ \{1\},\ \nu^{(5)})$	crush level 3, peel ring 1	$+1$

These are all distinct elements of the ground set, with different signs and generally different total weights. The ground set $\widetilde{\mathcal{G}}_{n,c}$ is the space of *all possible compressions* of all CPPs with max entry up to n.

The conjecture $Q_n \in \mathbb{N}[q]$ therefore says: for every CPP and every odd-depth compression of it (with $|S| + |U|$ odd), there exists a matching even-depth compression at the same total weight. The involution's job is to find these matchings systematically.

Remark 14 (S, U, and the ALCD)**.** The partial disassembly picture should not be confused with the ALCD. The **ALCD** $\mathfrak{d} = (m_k^{(j)})$ is the growth diagram's native representation: T face labels at each of n rings, equivalent to the CPP via the growth diagram bijection (Section 3). It carries *no sign* and records the complete internal structure.

The components S and U are **sign-carrying** objects from the algebraic expansion of the Euler products in (5): S carries sign $(-1)^{|S|}$, U carries sign $(-1)^{|U|}$, and the CPP ν carries sign $+1$. When the involution removes a ring (into S), the ALCD loses a full row of T face labels and S gains one number. When the involution extracts weight (into U), the ALCD's face labels at ring i are reduced and U gains one number. In both cases, detailed structure is compressed into a single scalar.

4.2 Compression of a CPP

The ground set has a natural interpretation: each triple (S, U, ν) represents a **compression** of a cylindric plane partition. A compression selects certain rings to crush or mark, producing a smaller, lighter object that still records all the weight.

Definition 15 (Compression)**.** Let $\tilde{\nu}$ be a CPP with max entry n. A **compression** of $\tilde{\nu}$ is a triple (S, U, ν) obtained in two stages:

1. **Structural compression** (choosing S): select a subset $S \subseteq \{1, \ldots, n\}$ of levels to crush. Crushing level k applies the inverse Burge rule at all T faces of ring k, merging levels $k-1$ and k. After crushing all levels in S, the residual CPP has max entry $m = n - |S|$. Each crushed level contributes sign -1 and weight k (the level index).

2. **Weight compression** (choosing U): select a subset $U \subseteq \{1, \ldots, m\}$ of remaining rings to mark. Marking ring i extracts i units of face-label weight from the growth diagram at ring i, storing them as the Euler mark $i \in U$. The CPP ν is the residual after both stages. Each mark contributes sign -1 and weight i.

The total sign is $(-1)^{|S|+|U|}$ and the total weight is $\sigma(S) + \sigma(U) + |\nu|$.

At one extreme ($S = U = \emptyset$), no compression occurs and the triple is simply $(\emptyset, \emptyset, \tilde{\nu})$ with sign $+1$. At the other extreme ($|S| = n$), every level is crushed, the CPP is empty, and all weight is carried by $\sigma(S)$.

Remark 16 (The ground set as a space of compressions)**.** A single CPP $\tilde{\nu}$ with max entry n generates 2^n compressions (one for each choice of S), and each residual CPP generates further choices of U. Different compressions of the same CPP are distinct elements of the ground set with (generally) different signs and total weights.

The full ground set $\widetilde{\mathcal{G}}_{n,c}$ is the union of all compressions of all CPPs with max entry up to n. The conjecture $Q_n \in \mathbb{N}[q]$ asserts that across all CPPs and all their compressions, the positive-signed terms outweigh the negative-signed ones at every power of q. The involution's task is to pair each negative compression with a positive one at the same total weight.

4.3 Two levels of compression

The ground set (7) can be compressed by absorbing the internal signs.

Level 2 (pairs). If $h_{m,c}(q) := (q;q)_m \cdot f_{m,c}(q) \in \mathbb{N}[q]$ (i.e., all coefficients are non-negative), then the (U, ν) data can be compressed into a single object τ counted by $h_{m,c}(q)$:

$$\mathcal{G}_{n,c} = \bigsqcup_{j=0}^{n} \{(S, \tau) : S \subseteq \{1, \ldots, n\},\ |S| = j,\ \tau \in \mathcal{H}_{n-j,c}\}, \tag{8}$$

with sign $(-1)^{|S|}$. The Euler signs from U cancel internally.

This compression works when $d \not\equiv 0 \pmod 3$ (verified for $d = 1, 2, 4, 5, 7, 8$). When $d \equiv 0 \pmod 3$, the polynomial $h_{m,c}(q)$ has negative coefficients, and the compression fails: one must work with the full triples (7).

Level 3 (triples). This is the universal description. Every component is manifestly non-negative: S and U are subsets (counted by 1), and ν is a CPP (counted by $f_{m,c}(q) \in \mathbb{N}[q]$). The signs come *only* from $(-1)^{|S|+|U|}$, never from the objects themselves. The non-negativity of h_m is irrelevant at this level.

4.4 Peel and glue ARE the involution

The central claim of this note is:

> *The sign-reversing involution on $\widetilde{\mathcal{G}}_{n,c}$ is not merely "implemented by" peel and glue. Every non-trivial move of the involution IS a peel or glue operation—there is no other mechanism available.*

To see why, consider what the involution must do. Suppose a map $\iota\colon \widetilde{\mathcal{G}}_{n,c} \to \widetilde{\mathcal{G}}_{n,c}$ exists that proves $Q_{n,c}(q) \in \mathbb{N}[q]$. Such a map is not yet constructed—its existence is equivalent to the conjecture—but we can deduce what form it must take. It acts on triples (S, U, ν) and must satisfy three constraints:

1. $\iota^2 = \mathrm{id}$ (involutive).

2. If $\iota(S,U,\nu) = (S',U',\nu') \neq (S,U,\nu)$, then $|S'|+|U'| \equiv |S|+|U|+1 \pmod 2$ (sign reversal).

3. $\sigma(S)+\sigma(U)+|\nu| = \sigma(S')+\sigma(U')+|\nu'|$ (weight preservation).

If such an ι exists, the fixed-point generating function gives the Q-polynomial:

$$Q_{n,c}(q) = \sum_{\substack{(S,U,\nu) \\ \iota(S,U,\nu)=(S,U,\nu)}} (-1)^{|S|+|U|}\, q^{\sigma(S)+\sigma(U)+|\nu|}. \tag{9}$$

We now show that ι, if it exists, must be a peel or a glue.

Sign reversal requires changing $|S|+|U|$ by ± 1. Since S and U are subsets (discrete objects), this means either adding or removing a single element from S or U. But adding or removing an element changes the weight, so the CPP ν must compensate. The *only* way to adjust a CPP's weight while preserving its cylindric structure is to peel or glue a ring in the growth diagram. There is no "move a box" operation on CPPs that bypasses the growth diagram: the interlacing constraints between the three partitions (ν^1,ν^2,ν^3) are too rigid.

This means that ι is not something separate that "uses" peel and glue as subroutines. Each application of ι to a non-fixed-point triple IS exactly one peel or one glue:

$$\iota(S,U,\nu) = \begin{cases} \text{peel: strip a ring from } \nu\text{, store in } S \text{ or } U & \text{(triple gets "smaller")}, \\ \text{glue: take from } S \text{ or } U\text{, build a ring on } \nu & \text{(triple gets "bigger")}. \end{cases}$$

Every non-fixed-point pair $\{x,\iota(x)\}$ consists of one "big" triple and one "small" triple. The map ι sends the big one to the small one via peel, and the small one back to the big one via glue. Since peel and glue are mutual inverses (Proposition 5), $\iota^2 = \mathrm{id}$.

The entire involution is therefore encoded in a single **decision rule**: given a triple, does it get peeled or glued, and at which ring? This decision rule is the *canonical choice* of Section 4.5. The mechanism is given; the logic is missing.

The two fundamental moves are as follows.

Definition 17 (The two fundamental moves)**.** Given a triple (S,U,ν) with $m = n - |S|$, the involution performs *exactly one* of the following:

Move U (Euler peel/glue): trade a mark for weight. Change $|U|$ by ± 1 while keeping the CPP's max entry m fixed.

$$(S,U,\nu) \;\longleftrightarrow\; (S,\; U\triangle\{i\},\; \nu') \qquad \text{where } \max(\nu') = m,\ |\nu'| = |\nu| \mp i.$$

Peel direction (add a mark): Choose $i \in \{1,\dots,m\}\setminus U$. Redistribute weight i from ν to U: this means peeling cubes from the interior of the growth diagram (modifying face labels at ring i) to extract exactly i units of weight, then recording the mark $i \in U$. The CPP shrinks: $|\nu'| = |\nu| - i$.

Glue direction (remove a mark): Choose $i \in U$. Remove the mark and inject weight i back into ν: this means inserting face-label weight at ring i via the forward Burge rule, growing the CPP by i units.

Move S (crossing peel/glue): trade a level for a ring. Change $|S|$ by ± 1, which changes the CPP's max entry.

$$(S,U,\nu) \;\longleftrightarrow\; (S\triangle\{k\},\; U',\; \nu') \qquad \text{where } \max(\nu') = m \mp 1.$$

PEEL DIRECTION (add a crossing): Apply the inverse Burge rule (Section 2.2) to the *outermost ring* of ν, extracting face labels $(m_0, \ldots, m_{T-1})$. Record the level $k = m$ in S, and adjust U to $U' \subseteq \{1, \ldots, m-1\}$. The CPP loses its top ring: $\max(\nu') = m - 1$.

GLUE DIRECTION (remove a crossing): Take a level $k \in S$, construct face labels from the available weight budget, and apply the forward Burge rule (Section 2.1) to add a new outermost ring. The CPP gains a top ring: $\max(\nu') = m + 1$.

Remark 18 (Why only two moves, not three)**.** One might consider a third operation—**Move C** (cross-to-mark conversion)—that trades an element of S for an element of U. Since $|S|$ decreases by 1 and $|U|$ increases by 1, the total $|S| + |U|$ is unchanged, so the sign is preserved. Move C is therefore *sign-neutral*: it cannot serve as the involution step itself. It could in principle serve as a preparatory conjugation (apply Move C, then Move U, then undo Move C), but this composite operation reduces to a single Move S. The two fundamental moves suffice.

Remark 19 (Properties of peel and glue)**.** Both fundamental moves have three key properties.

Weight-preserving. Peel does not destroy weight—it *redistributes* it. By the weight identity (Proposition 3), a full-ring peel satisfies $|\nu| = |\nu'| + \sum_k m_k$: the CPP shrinks, and the extracted face labels carry the difference. In the involution context, that face-label weight flows into $\sigma(S)$ or $\sigma(U)$, so the total $\sigma(S) + \sigma(U) + |\nu|$ is preserved. Glue reverses the flow: weight moves from S or U back into the CPP.

Sign-reversing. Move U adds or removes an element from U, changing $|U|$ by ± 1. Move S adds or removes an element from S, changing $|S|$ by ± 1. Either way, the sign $(-1)^{|S|+|U|}$ flips. This is what makes the cancellation work: each peel–glue pair matches a positive triple with a negative triple of the same weight.

Mutually inverse (not self-inverse). Peel and glue are inverses of each other (Proposition 5): peel followed by glue with the same face labels returns the original CPP. But peel is *not* an involution—applying it twice strips two rings. The involution ι is built from this inverse pair: it applies peel in one direction and glue in the other, so that $\iota^2 = \mathrm{id}$.

4.5 The canonical choice

The preceding analysis shows that the *mechanism* of the involution is completely determined: it must be a peel or a glue. What remains open is the **canonical choice**: given a non-fixed-point triple (S, U, ν), which specific move does the involution apply?

The requirements on this choice are:

1. **Deterministic**: the choice depends only on (S, U, ν), with no auxiliary data.
2. **Self-inverse**: if the chosen move sends $(S, U, \nu) \mapsto (S', U', \nu')$, then the chosen move at (S', U', ν') must send it back to (S, U, ν).
3. **Weight-preserving**: the move must transfer *exactly* the right amount of weight between components.
4. **Well-defined on fixed points**: the triples where no canonical peel exists are precisely the fixed points, and their generating function must equal $Q_{n,c}(q)$.

Remark 20 (The peel/glue dichotomy)**.** For each non-fixed-point triple, the involution either peels (extracting structure from ν into S or U) or glues (injecting structure from S or U back into ν). The triple and its image form a *peel–glue pair*: one member is in the "peeled" state and the other

in the "glued" state. The canonical choice amounts to a consistent rule for deciding which state a given triple is in.

In Franklin's classical involution on Euler's pentagonal number theorem, the analogous choice is: "does the smallest part fit in the diagonal slope, or does the diagonal slope fit in the bottom row?" The involution has a single canonical comparison. For the cylindric case, the analogous comparison must account for three interlacing partitions wrapped around a cylinder—a fundamentally harder combinatorial problem.

Remark 21 (The asymmetry of peel and glue)**.** Peel and glue are mutual inverses (Proposition 5), but they are *not* symmetric in terms of information.

Peel is deterministic. Given a CPP ν, the inverse Burge rule at each face is a well-defined function: the side vertices α, β and outer vertex λ uniquely determine the inner vertex ν and face label m. A full-ring peel produces a unique smaller CPP ν' and unique face labels $(m_0, \ldots, m_{T-1})$. No choices are involved.

Glue requires data. The forward Burge rule is equally deterministic, but it takes the face labels as *input*. To glue a ring onto ν', one must specify $(m_0, \ldots, m_{T-1})$. Different face labels produce different CPPs.

This asymmetry has a direct consequence for the ground set. The triple (S, U, ν) does not encode enough information to reconstruct a unique "uncompressed" CPP. For the S component: the level index $k \in S$ records *which* ring was crushed, but not the T face labels that were there—those are lost. For the U component: the subset U and the CPP ν are algebraically independent objects (from different factors in the Q-polynomial expansion), so there is no literal extraction to reverse.

In the involution, this asymmetry concentrates the difficulty in the glue direction: peeling a non-fixed-point triple is automatic, but gluing its partner back requires choosing the right face labels. The canonical choice (Section 4.5) must resolve this.

Remark 22 (Contrast with the LGV involution)**.** In the classical Lindström–Gessel–Viennot setting, the sign-reversing involution on intersecting lattice path families is purely local: find the *first* intersection point of two paths, swap their tails beyond that point, and the crossing is resolved. The set of crossing points changes by exactly one element, the sign flips, and the weight is automatically preserved because the same edges are traversed (just by different paths).

One might hope that the crossing set $S \subseteq \{1, \ldots, n\}$ in our ground set plays the same role: S records "crossing levels" in the NILP tower, and the involution "uncrosses" them one at a time. This interpretation is morally correct—S does record which levels carry crossings—but the mechanism is fundamentally different.

The LGV paths live in the plane, and a tail-swap is a local operation that preserves all path constraints. Our paths live on a *cylinder*: three paths wrap around with cyclic interlacing $\nu^1 \trianglerighteq \nu^2 \trianglerighteq \nu^3 \trianglerighteq \nu^1$. Swapping tails at a single level would violate these interlacing conditions at other levels. Instead, "uncrossing level k" requires a *whole-ring* restructuring of the CPP via peel/glue: the inverse Burge rule is applied at all T faces around the circle, globally adjusting the path geometry to accommodate the change in S. The crossing set changes by one element (as in LGV), but the cost is a full ring operation (unlike LGV).

Remark 23 (Why: the determinant is missing)**.** The deeper reason the LGV involution fails is that the underlying generating function is not a determinant.

In the classical setting, the Jacobi–Trudi formula $s_\lambda = \det(h_{\lambda_i - i+j})$ expresses the Schur function as a determinant. The LGV lemma interprets this: the expansion $\det = \sum_\sigma \operatorname{sgn}(\sigma) \prod a_{i,\sigma(i)}$ runs over permutations σ, each corresponding to a path family. The sign-reversing involution (uncross,

swap tails) exploits the determinantal structure: swapping two rows of a determinant flips the sign, and that algebraic fact has a direct combinatorial counterpart.

For cylindric partitions, $\mathrm{GK}_c(z,q)$ is a *trace* of a product of transfer matrices around the cylinder, not a determinant. The cylindric Schur functions do not satisfy Jacobi–Trudi. The alternating-sum structure of $Q_n = \sum(-1)^{|S|+|U|}\, q^{\sigma(S)+\sigma(U)+|\nu|}$ arises instead from the Euler factors $(zq;q)_\infty$ and $(q;q)_m$ applied to this non-determinantal generating function (see the derivation in Section 4).

So the Q-polynomial has the *form* of a determinant expansion (a signed sum where massive cancellation is expected) without the *source* being a determinant. The involution must accomplish the same cancellation that the LGV involution achieves for genuine determinants, but without the algebraic shortcut that determinantal structure provides. Peel and glue supply the mechanism; the canonical choice (Section 4.5) must supply the logic.

4.6 Interior crushing

The standard peel always operates at the *boundary*: it removes the outermost ring (level m) of the NILP tower, crushing the top-level paths against the upper boundary. But there is no reason to restrict to the boundary.

Definition 24 (Interior crush)**.** Let ν be a CPP with max entry m, corresponding to a NILP tower with levels $1, 2, \ldots, m$. An **interior crush at level** k ($1 \le k \le m$) merges levels $k-1$ and k by applying the inverse Burge rule at all T faces of ring k: the face labels at ring k are extracted, and the two levels collapse into one. The result is a CPP ν' with max entry $m-1$, and a sequence of extracted face labels at ring k.

The standard peel (Section 3) is the special case $k = m$: crush at the top level. But crushing can occur at any interior level where two adjacent paths are close enough to merge.

The key advantage of interior crushing is that it resolves the **uniqueness problem** for S. When S records the *level indices* where crushes occurred, distinctness is automatic: each crush happens at a different level, so the elements of S are distinct by construction.

$$S = \{k_1, k_2, \ldots, k_j\} \textit{ means: levels } k_1, k_2, \ldots, k_j \textit{ have been crushed out of the tower.}$$

This gives a concrete geometric meaning to the elements of S that was missing from the top-peel picture. In top-peeling, $k \in S$ was an abstract index whose value had to be chosen (and might not match the face-label total). In interior crushing, $k \in S$ IS the level index—no choice required.

Remark 25 (Connection to LGV)**.** Interior crushing is closer to the LGV involution than top-peeling is. In LGV, you find crossings *wherever they occur* in the path family (not just at the boundary) and resolve them locally. Interior crushing similarly finds crushable pairs wherever they occur in the tower and resolves them at their natural level. The element $k \in S$ records *where* the crush happened, just as the LGV crossing set records where crossings occurred.

Remark 26 (Weight and the gap)**.** Crushing at level k extracts the face labels at ring k, with total weight $b_k = \sum_{i=0}^{T-1} m_i^{(k)}$. But the element $k \in S$ contributes weight k (the level index) to $\sigma(S)$, not b_k. If $b_k \neq k$, the difference $k - b_k$ must be absorbed by adjusting U or redistributing weight in ν'. Concretely:

$$\underbrace{k}_{\text{added to } \sigma(S)} \quad = \quad \underbrace{b_k}_{\text{extracted from } \nu} \quad + \quad \underbrace{(k - b_k)}_{\text{from } U \text{ adjustment}} \quad .$$

When $k = b_k$ (the face-label total at ring k happens to equal the level index), the crush is *clean*: no U-adjustment is needed. Otherwise, the difference must be compensated, giving a coupling between the crush and the Euler marks.

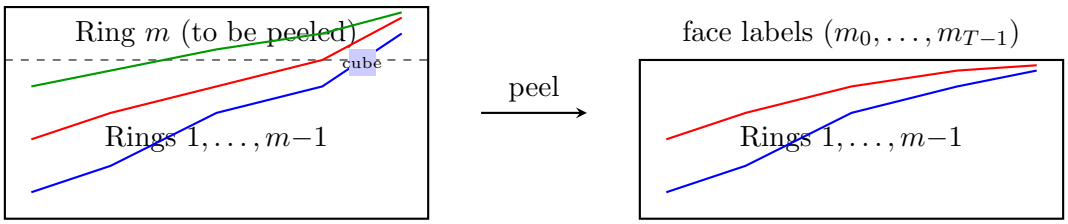
5 Peel and glue in NILP terms

In the non-intersecting lattice path model, a CPP ν with max entry m corresponds to a family of paths on the cylinder, arranged in a tower of m levels. Each level has $T = d$ steps around the cylinder, and the paths maintain their non-intersecting order at every step.

5.1 Peeling a ring = removing the top level of the tower

Peeling ring m (the outermost) from the NILP tower means:

1. At each step k around the cylinder, examine how the paths at level m relate to the paths at level $m-1$.
2. If both the left path and right path at step k "pinch" (both take up-steps beyond level $m-1$), a cascade occurs: the pinch at column i propagates to the next free column.
3. The face label m_k counts how many cascade chains reach column 0 — the number of cubes at this step.
4. After removing the top level, the remaining paths form a valid NILP tower of height $m-1$.



5.2 Gluing a ring = adding a new top level

Gluing reverses the peel: given the $m-1$ level tower and face labels $(m_0, \ldots, m_{T-1})$, the forward Burge rule reconstructs the top-level paths. The face labels determine where the new top-level paths go: each cube (face label unit) inserts a new path-step, extending the paths upward.

In terms of the involution on triples (S, U, ν), the two fundamental moves (Definition 17) have transparent NILP interpretations:

- **Move S (peel direction):** Remove the top level of the NILP tower: collapse the three paths at level m into the crossing marker $m \in S$. The extracted face labels encode the path geometry at the removed level. The tower shrinks: $m \to m-1$.

  Move S (glue direction): Restore a level: take $k \in S$, reconstruct face labels from the weight budget, and apply the forward rule to build new top-level paths. The tower grows: $m \to m+1$.

- **Move U (peel direction):** The tower keeps the same height m. Instead, modify the face labels at ring i *within* the tower, extracting i units of weight and storing them as the Euler mark $i \in U$. In NILP terms: remove i cubes from the interior of the path family (tighten the gaps between paths at specific steps).

  Move U (glue direction): Take the mark $i \in U$ and inject i cubes back into the interior (widen the gaps between paths at specific steps).

Every involution step—without exception—is either a peel (extracting structure from the tower into S or U) or a glue (injecting structure from S or U back into the tower). The canonical choice of which to perform at each non-fixed-point triple is the open problem (Section 4.5).

5.3 Compression = a partially disassembled tower

The two operations combine to give a NILP picture of compression (Definition 15). A triple (S, U, ν) with $m = n - |S|$ represents a tower that has been **partially disassembled** in two ways:

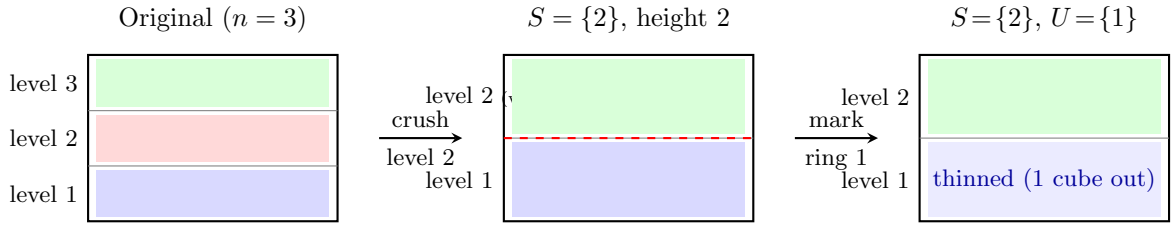
1. **Floors removed** (structural compression, S). Each $k \in S$ records a level that has been crushed out of the tower. The three paths at that level are gone; the levels above have collapsed down. The tower shrinks from n levels to m levels.

   Concretely: in a tower of height $n = 3$ with $S = \{2\}$, the second floor is removed and the third floor becomes the new second. The cubes that filled the gap between levels 1 and 2 are extracted as face labels, then discarded (their total weight b_2 is absorbed into $\sigma(S)$).

2. **Floors thinned** (weight compression, U). Each $i \in U$ records a level where cubes have been extracted from the interior. The paths at level i remain, but the gap between levels $i-1$ and i has been tightened: exactly i cubes are missing compared to the uncompressed tower.

   Concretely: if $U = \{2\}$ in a tower of height $m = 3$, then 2 cubes have been removed from ring 2—the paths at levels 1 and 2 are closer together at two of the T steps around the cylinder.

The residual CPP ν is whatever remains: a valid NILP tower of height m, with thinner gaps at the marked rings.



The two compressions differ in their effect on the tower:

	Structural (S)	**Weight (U)**
Tower height	decreases by 1	unchanged
Paths at that level	removed entirely	remain, gaps tightened
Cubes affected	all cubes at one level	exactly i cubes thinned
Reversal	glue: needs face labels	glue: re-insert i cubes

To **reconstruct** the original n-level tower from (S, U, ν): first undo the U-marks (inject cubes back into the thinned rings), then undo the S-crushes (glue the removed floors back, from bottom to top).

6 The $n = 1$ case: injection as cube addition

For $n = 1$, the ground set simplifies dramatically. There are only three types of triples:

Triple	**Sign**	**Weight**	**Meaning**
$(\emptyset, \emptyset, \nu)$, $\max(\nu) = 1$	$+1$	$\lvert\nu\rvert$	unmarked NILP family
$(\emptyset, \{1\}, \nu)$, $\max(\nu) = 1$	-1	$1 + \lvert\nu\rvert$	Euler-marked NILP family
$(\{1\}, \emptyset, \nu_\emptyset)$, $\max(\nu_\emptyset) = 0$	-1	1	crossed (empty CPP)

The involution must pair each negative triple with a distinct positive triple of the same weight. At weight w:

- Positive triples: CPPs with max $= 1$ and weight w. Count: $f_1(w) := [q^w]\, f_{1,c}(q)$.
- Negative triples:
  - Euler-marked CPPs with weight $w - 1$. Count: $f_1(w - 1)$.
  - One crossed triple (only at $w = 1$).

For $w \geq 2$, the involution reduces to an **injection**:

$$\varphi\colon \{\text{CPPs with } \max = 1,\ |\nu| = w - 1\} \hookrightarrow \{\text{CPPs with } \max = 1,\ |\nu| = w\}. \tag{10}$$

The image of φ pairs with the Euler-marked triples; the non-images are the fixed points of the involution, contributing to $Q_{1,c}(q)$.

In NILP terms, φ adds one cube to the NILP family:

$$\varphi(\nu) = \nu + \text{one cube at a canonical position.}$$

Each CPP $\nu = (\nu^1, \nu^2, \nu^3)$ with max $= 1$ has partitions $\nu^i = (1^{a_i})$. Adding a cube means appending a row to one of the three partitions: $a_k \to a_k + 1$ for some $k \in \{1, 2, 3\}$.

The **slack vector** $s = (s_1, s_2, s_3)$, where $s_k = a_k - a_{k-1 \bmod 3} + c_{k-1 \bmod 3}$ measures the room available for each path, determines which paths can be extended: path k can receive a cube if and only if $s_k > 0$.

7 The $d \equiv 0 \pmod 3$ obstruction

For $d \equiv 0 \pmod 3$, the CPP weight distribution at max $= 1$ oscillates with period 3:

$$f_1(w) = \begin{cases} d/3 + 1 & \text{if } w \equiv 0 \pmod 3, \\ d/3 & \text{otherwise.} \end{cases} \qquad \text{(for } w \text{ sufficiently large).}$$

The extra CPP at weights $w \equiv 0 \pmod 3$ is the *balanced* configuration $\nu = (1^a, 1^a, 1^a)$ with slack (a, a, a) (equal room in all three directions).

The injection (10) maps CPPs of weight w into CPPs of weight $w + 1$. At weights $w \equiv 0 \pmod 3$ (where there are $d/3 + 1$ CPPs) mapping to $w + 1 \equiv 1 \pmod 3$ (where there are only $d/3$ CPPs), the injection has a **deficit**: more sources than targets.

This deficit is exactly why $Q_{1,c}(q)$ has negative coefficients when computed with $(q)_1 = 1 - q$. The conjecture resolves this by using $(q^\ell; q^\ell)_n$ with $\ell = \gcd(d, 3) = 3$: the factor $(1 - q^3)$ has the correct periodicity to match the period-3 oscillation, absorbing the deficit.

8 Summary

Operation	**Growth diagram**	**NILP tower**
Peel (inverse rule)	Remove outermost ring; extract face labels	Remove top level of tower; extract cubes as face labels
Glue (forward rule)	Add outermost ring; insert face labels	Add top level to tower; insert cubes from face labels
Move S (crossing)	Peel/glue ring m; trade level $\leftrightarrow S$	Collapse/expand top-level paths $\leftrightarrow$ crossing marker
Move U (Euler mark)	Redistribute weight at ring i; trade weight $\leftrightarrow U$	Tighten/widen gaps at ring i; trade cubes $\leftrightarrow$ Euler mark

Move C (cross-to-mark conversion) is sign-neutral and does not appear as a standalone involution step (Remark 18).

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